

DATASHEET / LIST

ISO BOM / MTO Generator

Aggregates BOM/MTO from E3D · SP3D · PDS isometrics and support drawings, exporting to Excel. Unified RUNALL + shared UI theme.

Key Purpose

Why this program

1. One tool for every isometric source

– E3D ISO, SP3D ISO, PDS ISO and Support drawings all read by a single picker

→ **Effect:** No need to remember or launch four separate BOM scripts — one window covers the whole plant regardless of CAD origin

2. Batch DXF/DWG to Excel BOM in one click

– Select many drawings, choose the BOM type, press Apply — the matching extractor runs in-memory on the whole batch

→ **Effect:** Material take-off that used to be manual counting becomes a few seconds of automated aggregation, with no transcription error

3. Results ready to open on the spot

– Generated .xlsx files are detected automatically and listed; select one and press Open

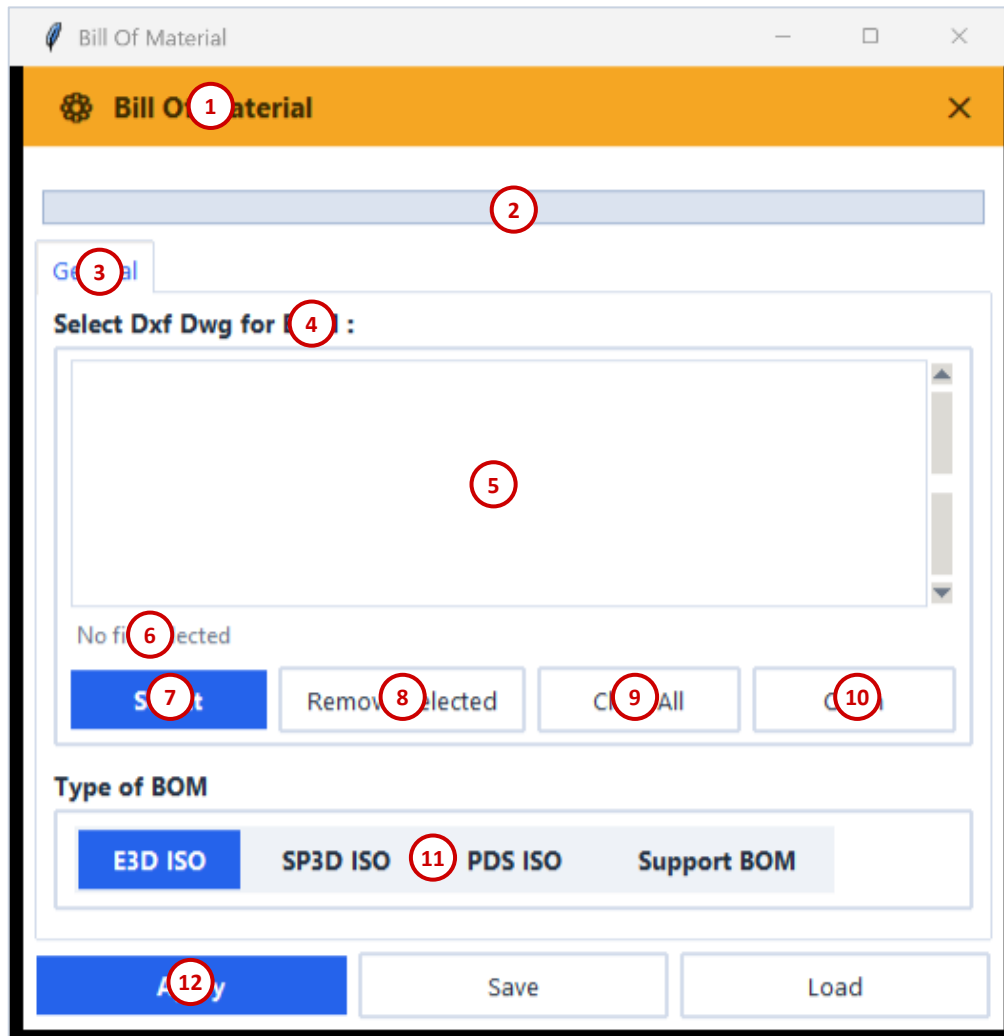
→ **Effect:** The engineer sees and opens the finished BOM without hunting through folders

4. Reusable option sets

Save / Load remembers the chosen files and BOM type

Required inputs : One or more isometric / support DXF (DWG) drawings + a BOM type (E3D / SP3D / PDS / Support)

Main Window – Item Guide



1. Title bar — Bill Of Material
2. Progress bar (batch status)
3. General tab
4. Select Dxf Dwg for BOM — section
5. Selected drawings / generated files list
6. File count status (No file selected)
7. Select — add DXF drawings
8. Remove Selected
9. Clear All
10. Open — open highlighted file
11. Type of BOM — E3D / SP3D / PDS / Support
12. Apply — run BOM; Save / Load options

Overview

What it does

1. Aggregates BOM/MTO from E3D
2. SP3D
3. PDS isometrics and support drawings
4. Exporting to Excel
5. Unified RUNALL
6. Shared UI theme

Category: Datasheet / List

Folder: MTO

Main scripts: bom_runall.py

Scripts / Status: 6 scripts · Operational

Tags: MTO · BOM · ISO · Excel

Output Samples (1/6)

ISO BOM AUTO GEN FROM ISO DWG - MTO Data

ISO BOM AUTO GEN FROM ISO DWG - TOTAL BOM

Output Samples

Output Samples (2/6)

ISO Dwg No	Item No	ISO Description	Item No	Item	Size	Material	Description	Qty	Volume	Weight	Unit	Unit Price	Cost	Unit Price	Cost
110110-01-01	1	ISO 110110-01-01	1	ISO 110110-01-01	1	ISO 110110-01-01	ISO 110110-01-01	1	1.00	1.00	1	1.00	1.00	1.00	1.00
110110-01-02	2	ISO 110110-01-02	2	ISO 110110-01-02	2	ISO 110110-01-02	ISO 110110-01-02	2	2.00	2.00	2	2.00	2.00	2.00	2.00
110110-01-03	3	ISO 110110-01-03	3	ISO 110110-01-03	3	ISO 110110-01-03	ISO 110110-01-03	3	3.00	3.00	3	3.00	3.00	3.00	3.00
110110-01-04	4	ISO 110110-01-04	4	ISO 110110-01-04	4	ISO 110110-01-04	ISO 110110-01-04	4	4.00	4.00	4	4.00	4.00	4.00	4.00
110110-01-05	5	ISO 110110-01-05	5	ISO 110110-01-05	5	ISO 110110-01-05	ISO 110110-01-05	5	5.00	5.00	5	5.00	5.00	5.00	5.00
110110-01-06	6	ISO 110110-01-06	6	ISO 110110-01-06	6	ISO 110110-01-06	ISO 110110-01-06	6	6.00	6.00	6	6.00	6.00	6.00	6.00
110110-01-07	7	ISO 110110-01-07	7	ISO 110110-01-07	7	ISO 110110-01-07	ISO 110110-01-07	7	7.00	7.00	7	7.00	7.00	7.00	7.00
110110-01-08	8	ISO 110110-01-08	8	ISO 110110-01-08	8	ISO 110110-01-08	ISO 110110-01-08	8	8.00	8.00	8	8.00	8.00	8.00	8.00
110110-01-09	9	ISO 110110-01-09	9	ISO 110110-01-09	9	ISO 110110-01-09	ISO 110110-01-09	9	9.00	9.00	9	9.00	9.00	9.00	9.00
110110-01-10	10	ISO 110110-01-10	10	ISO 110110-01-10	10	ISO 110110-01-10	ISO 110110-01-10	10	10.00	10.00	10	10.00	10.00	10.00	10.00
110110-01-11	11	ISO 110110-01-11	11	ISO 110110-01-11	11	ISO 110110-01-11	ISO 110110-01-11	11	11.00	11.00	11	11.00	11.00	11.00	11.00
110110-01-12	12	ISO 110110-01-12	12	ISO 110110-01-12	12	ISO 110110-01-12	ISO 110110-01-12	12	12.00	12.00	12	12.00	12.00	12.00	12.00
110110-01-13	13	ISO 110110-01-13	13	ISO 110110-01-13	13	ISO 110110-01-13	ISO 110110-01-13	13	13.00	13.00	13	13.00	13.00	13.00	13.00
110110-01-14	14	ISO 110110-01-14	14	ISO 110110-01-14	14	ISO 110110-01-14	ISO 110110-01-14	14	14.00	14.00	14	14.00	14.00	14.00	14.00
110110-01-15	15	ISO 110110-01-15	15	ISO 110110-01-15	15	ISO 110110-01-15	ISO 110110-01-15	15	15.00	15.00	15	15.00	15.00	15.00	15.00
110110-01-16	16	ISO 110110-01-16	16	ISO 110110-01-16	16	ISO 110110-01-16	ISO 110110-01-16	16	16.00	16.00	16	16.00	16.00	16.00	16.00
110110-01-17	17	ISO 110110-01-17	17	ISO 110110-01-17	17	ISO 110110-01-17	ISO 110110-01-17	17	17.00	17.00	17	17.00	17.00	17.00	17.00
110110-01-18	18	ISO 110110-01-18	18	ISO 110110-01-18	18	ISO 110110-01-18	ISO 110110-01-18	18	18.00	18.00	18	18.00	18.00	18.00	18.00
110110-01-19	19	ISO 110110-01-19	19	ISO 110110-01-19	19	ISO 110110-01-19	ISO 110110-01-19	19	19.00	19.00	19	19.00	19.00	19.00	19.00
110110-01-20	20	ISO 110110-01-20	20	ISO 110110-01-20	20	ISO 110110-01-20	ISO 110110-01-20	20	20.00	20.00	20	20.00	20.00	20.00	20.00
110110-01-21	21	ISO 110110-01-21	21	ISO 110110-01-21	21	ISO 110110-01-21	ISO 110110-01-21	21	21.00	21.00	21	21.00	21.00	21.00	21.00
110110-01-22	22	ISO 110110-01-22	22	ISO 110110-01-22	22	ISO 110110-01-22	ISO 110110-01-22	22	22.00	22.00	22	22.00	22.00	22.00	22.00
110110-01-23	23	ISO 110110-01-23	23	ISO 110110-01-23	23	ISO 110110-01-23	ISO 110110-01-23	23	23.00	23.00	23	23.00	23.00	23.00	23.00
110110-01-24	24	ISO 110110-01-24	24	ISO 110110-01-24	24	ISO 110110-01-24	ISO 110110-01-24	24	24.00	24.00	24	24.00	24.00	24.00	24.00
110110-01-25	25	ISO 110110-01-25	25	ISO 110110-01-25	25	ISO 110110-01-25	ISO 110110-01-25	25	25.00	25.00	25	25.00	25.00	25.00	25.00
110110-01-26	26	ISO 110110-01-26	26	ISO 110110-01-26	26	ISO 110110-01-26	ISO 110110-01-26	26	26.00	26.00	26	26.00	26.00	26.00	26.00
110110-01-27	27	ISO 110110-01-27	27	ISO 110110-01-27	27	ISO 110110-01-27	ISO 110110-01-27	27	27.00	27.00	27	27.00	27.00	27.00	27.00
110110-01-28	28	ISO 110110-01-28	28	ISO 110110-01-28	28	ISO 110110-01-28	ISO 110110-01-28	28	28.00	28.00	28	28.00	28.00	28.00	28.00
110110-01-29	29	ISO 110110-01-29	29	ISO 110110-01-29	29	ISO 110110-01-29	ISO 110110-01-29	29	29.00	29.00	29	29.00	29.00	29.00	29.00
110110-01-30	30	ISO 110110-01-30	30	ISO 110110-01-30	30	ISO 110110-01-30	ISO 110110-01-30	30	30.00	30.00	30	30.00	30.00	30.00	30.00
110110-01-31	31	ISO 110110-01-31	31	ISO 110110-01-31	31	ISO 110110-01-31	ISO 110110-01-31	31	31.00	31.00	31	31.00	31.00	31.00	31.00
110110-01-32	32	ISO 110110-01-32	32	ISO 110110-01-32	32	ISO 110110-01-32	ISO 110110-01-32	32	32.00	32.00	32	32.00	32.00	32.00	32.00
110110-01-33	33	ISO 110110-01-33	33	ISO 110110-01-33	33	ISO 110110-01-33	ISO 110110-01-33	33	33.00	33.00	33	33.00	33.00	33.00	33.00
110110-01-34	34	ISO 110110-01-34	34	ISO 110110-01-34	34	ISO 110110-01-34	ISO 110110-01-34	34	34.00	34.00	34	34.00	34.00	34.00	34.00
110110-01-35	35	ISO 110110-01-35	35	ISO 110110-01-35	35	ISO 110110-01-35	ISO 110110-01-35	35	35.00	35.00	35	35.00	35.00	35.00	35.00
110110-01-36	36	ISO 110110-01-36	36	ISO 110110-01-36	36	ISO 110110-01-36	ISO 110110-01-36	36	36.00	36.00	36	36.00	36.00	36.00	36.00
110110-01-37	37	ISO 110110-01-37	37	ISO 110110-01-37	37	ISO 110110-01-37	ISO 110110-01-37	37	37.00	37.00	37	37.00	37.00	37.00	37.00
110110-01-38	38	ISO 110110-01-38	38	ISO 110110-01-38	38	ISO 110110-01-38	ISO 110110-01-38	38	38.00	38.00	38	38.00	38.00	38.00	38.00
110110-01-39	39	ISO 110110-01-39	39	ISO 110110-01-39	39	ISO 110110-01-39	ISO 110110-01-39	39	39.00	39.00	39	39.00	39.00	39.00	39.00
110110-01-40	40	ISO 110110-01-40	40	ISO 110110-01-40	40	ISO 110110-01-40	ISO 110110-01-40	40	40.00	40.00	40	40.00	40.00	40.00	40.00
110110-01-41	41	ISO 110110-01-41	41	ISO 110110-01-41	41	ISO 110110-01-41	ISO 110110-01-41	41	41.00	41.00	41	41.00	41.00	41.00	41.00
110110-01-42	42	ISO 110110-01-42	42	ISO 110110-01-42	42	ISO 110110-01-42	ISO 110110-01-42	42	42.00	42.00	42	42.00	42.00	42.00	42.00
110110-01-43	43	ISO 110110-01-43	43	ISO 110110-01-43	43	ISO 110110-01-43	ISO 110110-01-43	43	43.00	43.00	43	43.00	43.00	43.00	43.00
110110-01-44	44	ISO 110110-01-44	44	ISO 110110-01-44	44	ISO 110110-01-44	ISO 110110-01-44	44	44.00	44.00	44	44.00	44.00	44.00	44.00
110110-01-45	45	ISO 110110-01-45	45	ISO 110110-01-45	45	ISO 110110-01-45	ISO 110110-01-45	45	45.00	45.00	45	45.00	45.00	45.00	45.00
110110-01-46	46	ISO 110110-01-46	46	ISO 110110-01-46	46	ISO 110110-01-46	ISO 110110-01-46	46	46.00	46.00	46	46.00	46.00	46.00	46.00
110110-01-47	47	ISO 110110-01-47	47	ISO 110110-01-47	47	ISO 110110-01-47	ISO 110110-01-47	47	47.00	47.00	47	47.00	47.00	47.00	47.00
110110-01-48	48	ISO 110110-01-48	48	ISO 110110-01-48	48	ISO 110110-01-48	ISO 110110-01-48	48	48.00	48.00	48	48.00	48.00	48.00	48.00
110110-01-49	49	ISO 110110-01-49	49	ISO 110110-01-49	49	ISO 110110-01-49	ISO 110110-01-49	49	49.00	49.00	49	49.00	49.00	49.00	49.00
110110-01-50	50	ISO 110110-01-50	50	ISO 110110-01-50	50	ISO 110110-01-50	ISO 110110-01-50	50	50.00	50.00	50	50.00	50.00	50.00	50.00
110110-01-51	51	ISO 110110-01-51	51	ISO 110110-01-51	51	ISO 110110-01-51	ISO 110110-01-51	51	51.00	51.00	51	51.00	51.00	51.00	51.00
110110-01-52	52	ISO 110110-01-52	52	ISO 110110-01-52	52	ISO 110110-01-52	ISO 110110-01-52	52	52.00	52.00	52	52.00	52.00	52.00	52.00
110110-01-53	53	ISO 110110-01-53	53	ISO 110110-01-53	53	ISO 110110-01-53	ISO 110110-01-53	53	53.00	53.00	53	53.00	53.00	53.00	53.00
110110-01-54	54	ISO 110110-01-54	54	ISO 110110-01-54	54	ISO 110110-01-54	ISO 110110-01-54	54	54.00	54.00	54	54.00	54.00	54.00	54.00
110110-01-55	55	ISO 110110-01-55	55	ISO 110110-01-55	55	ISO 110110-01-55	ISO 110110-01-55	55	55.00	55.00	55	55.00	55.00	55.00	55.00
110110-01-56	56	ISO 110110-01-56	56	ISO 110110-01-56	56	ISO 110110-01-56	ISO 110110-01-56	56	56.00	56.00	56	56.00	56.00	56.00	56.00
110110-01-57	57	ISO 110110-01-57	57	ISO 110110-01-57	57	ISO 110110-01-57	ISO 110110-01-57	57	57.00	57.00	57	57.00	57.00	57.00	57.00
110110-01-58	58	ISO 110110-01-58	58	ISO 110110-01-58	58	ISO 110110-01-58	ISO 110110-01-58	58	58.00	58.00	58	58.00	58.00	58.00	58.00
110110-01-59	59	ISO 110110-01-59	59	ISO 110110-01-59	59	ISO 110110-01-59	ISO 110110-01-59	59	59.00	59.00	59	59.00	59.00	59.00	59.00
110110-01-60	60	ISO 110110-01-60	60	ISO 110110-01-60	60	ISO 110110-01-60	ISO 110110-01-60	60	60.00	60.00	60	60.00	60.00	60.00	60.00
110110-01-61	61	ISO 110110-01-61	61	ISO 110110-01-61	61	ISO 110110-01-61	ISO 110110-01-61	61	61.00	61.00	61	61.00	61.00	61.00	61.00
110110-01-62	62	ISO 110110-01-62	62	ISO 110110-01-62	62	ISO 110110-01-62	ISO 110110-01-62	62	62.00	62.00	62	62.00	62.00	62.00	62.00
110110-01-63	63	ISO 110110-01-63	63	ISO 110110-01-63	63	ISO 110110-01-63	ISO 110110-01-63	63	63.00	63.00	63	63.00	63.00	63.00	63.00
110110-01-64	64	ISO 110110-01-64	64	ISO 110110-01-64	64	ISO 110110-01-64	ISO 110110-01-64	64	64.00	64.00	64	64.00	64.00	64	

Output Samples (3/6)

[illegible]

ISO DWG NO	LINE NO	ISO DESCRIPTION	ITEM NO	ITEM DESCRIPTION	SIZE	MATERIAL	DESCRIPTION	Q'TY	UNIT	W'T	INSUL THK	REMARKS
2-1310-2F-186-001	110002-1110-011-480-250	"BFF" A" S" Suction Line	6	CHECK/DRAWING	1/2"	A156-WCB	BP #150	1	EA	-	-	
2-1310-2F-186-021	110002-1110-010-480-250	"BFF" A" S" Suction Line	7	MON	2"	A156-WCB	BP #152 D 15W 401A	1	EA	250	0	30
2-1310-2F-186-273	110002-1110-011-480-150	CLOSED COLD WATER SUPPLY LINE	7	GLOBE VAL	1/2"	A156-WCB	COA	1	EA	-	-	2050

ISO BOM AUTO GEN FROM ISO DWG - FITTING

ISO BOM AUTO GEN FROM ISO DWG - INST

Output Samples (4/6)

ISO 22716 No.	LINE NO.	DESCRIPTION	WELD NO.	DISPOTED	WELD SIZE	WELD TYPE	WELD MATERIAL	WELD TIME	MELTING DATE	TYPE	GRADE TYPE	TYPE	PREPARE	PRINT DATE	UNIT
	1	15510-15510-001-001-001	1	new	250	new	SA-1018-A	10	new	V-Groove	1-1/8"	1-1/8"	1-1/8"	new	100%
	2	15510-15510-001-001-002	2	new	250	new	SA-1018-A	10	new	V-Groove	1-1/8"	1-1/8"	1-1/8"	new	100%
	3	15510-15510-001-001-003	3	new	250	new	SA-1018-A	10	new	V-Groove	1-1/8"	1-1/8"	1-1/8"	new	100%
	4	15510-15510-001-001-004	4	new	250	new	SA-1018-A	10	new	V-Groove	1-1/8"	1-1/8"	1-1/8"	new	100%
	5	15510-15510-001-001-005	5	new	250	new	SA-1018-A	10	new	V-Groove	1-1/8"	1-1/8"	1-1/8"	new	100%
	6	15510-15510-001-001-006	6	new	250	new	SA-1018-A	10	new	V-Groove	1-1/8"	1-1/8"	1-1/8"	new	100%
	7	15510-15510-001-001-007	7	new	250	new	SA-1018-A	10	new	V-Groove	1-1/8"	1-1/8"	1-1/8"	new	100%
	8	15510-15510-001-001-008	8	new	250	new	SA-1018-A	10	new	V-Groove	1-1/8"	1-1/8"	1-1/8"	new	100%
	9	15510-15510-001-001-009	9	new	250	new	SA-1018-A	10	new	V-Groove	1-1/8"	1-1/8"	1-1/8"	new	100%
	10	15510-15510-001-001-010	10	new	250	new	SA-1018-A	10	new	V-Groove	1-1/8"	1-1/8"	1-1/8"	new	100%
	11	15510-15510-001-001-011	11	new	250	new	SA-1018-A	10	new	V-Groove	1-1/8"	1-1/8"	1-1/8"	new	100%
	12	15510-15510-001-001-012	12	new	250	new	SA-1018-A	10	new	V-Groove	1-1/8"	1-1/8"	1-1/8"	new	100%
	13	15510-15510-001-001-013	13	new	250	new	SA-1018-A	10	new	V-Groove	1-1/8"	1-1/8"	1-1/8"	new	100%
	14	15510-15510-001-001-014	14	new	250	new	SA-1018-A	10	new	V-Groove	1-1/8"	1-1/8"	1-1/8"	new	100%
	15	15510-15510-001-001-015	15	new	250	new	SA-1018-A	10	new	V-Groove	1-1/8"	1-1/8"	1-1/8"	new	100%
	16	15510-15510-001-001-016	16	new	250	new	SA-1018-A	10	new	V-Groove	1-1/8"	1-1/8"	1-1/8"	new	100%
	17	15510-15510-001-001-017	17	new	250	new	SA-1018-A	10	new	V-Groove	1-1/8"	1-1/8"	1-1/8"	new	100%
	18	15510-15510-001-001-018	18	new	250	new	SA-1018-A	10	new	V-Groove	1-1/8"	1-1/8"	1-1/8"	new	100%
	19	15510-15510-001-001-019	19	new	250	new	SA-1018-A	10	new	V-Groove	1-1/8"	1-1/8"	1-1/8"	new	100%
	20	15510-15510-001-001-020	20	new	250	new	SA-1018-A	10	new	V-Groove	1-1/8"	1-1/8"	1-1/8"	new	100%
	21	15510-15510-001-001-021	21	new	250	new	SA-1018-A	10	new	V-Groove	1-1/8"	1-1/8"	1-1/8"	new	100%
	22	15510-15510-001-001-022	22	new	250	new	SA-1018-A	10	new	V-Groove	1-1/8"	1-1/8"	1-1/8"	new	100%
	23	15510-15510-001-001-023	23	new	250	new	SA-1018-A	10	new	V-Groove	1-1/8"	1-1/8"	1-1/8"	new	100%
	24	15510-15510-001-001-024	24	new	250	new	SA-1018-A	10	new	V-Groove	1-1/8"	1-1/8"	1-1/8"	new	100%
	25	15510-15510-001-001-025	25	new	250	new	SA-1018-A	10	new	V-Groove	1-1/8"	1-1/8"	1-1/8"	new	100%
	26	15510-15510-001-001-026	26	new	250	new	SA-1018-A	10	new	V-Groove	1-1/8"	1-1/8"	1-1/8"	new	100%
	27	15510-15510-001-001-027	27	new	250	new	SA-1018-A	10	new						

WELDING BOM AUTO GEN FROM ISO DWG - FW

Output Samples (5/6)

WELDING BOM AUTO GEN FROM ISO DWG - SW

WELDING BOM AUTO GEN FROM ISO DWG - SW TOTAL

Output Samples (6/6)

SHOP/FLD	WELD BORE	WELD TYPE	WELD MATERIAL	WELD THK	WELDING DIA	WELD POINT	용작량 TOTAL	구입보수량 TOTAL	용작재료
FW	100	BW	A53-B	ERW BE SCH40 SCH80(≒3000)	8.00	2	0.10kg	0.16kg	E7018
FW	250	BW	SA106-B	SMLS BE SCH40	130.00	13	1.43kg	2.34kg	E7018
TOTAL	-	-	-	-	138.00	15	1.53kg	2.50kg	-

[illegible]

WELDING BOM AUTO GEN FROM ISO DWG - FW TOTAL

WELDING BOM AUTO GEN FROM ISO DWG - MAT CS

Notes & Information

- Part of the in-house Plant Engineering automation suite — category: Datasheet / List.
- Runs offline as a pure-Python / Tkinter tool; portable across PCs via OneDrive sync (needs Python + packages).
- Launch from the PC launcher (런처.bat) or run the main script: bom_runall.py.
- Output samples and the program window above reflect the current build.

Thank You

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